
Closing Evidence-Blindness in Radiology Report Generators via Dual-Adversarial Causal-Faithfulness Training

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Abstract

Vision-language models that draft radiology reports often produce text that does not reflect the underlying chest X-ray. Diagnostic studies have documented this evidence-blindness, and existing fixes wrap the trained model in inference-time filters or conformal selection without changing what the model has learned. We intervene at training time. We present dual-adversarial causal-faithfulness training, a LoRA fine-tune of MedGemma-4B-IT with two ingredients: a symmetric adversarial filter that downweights training rows that either a text-only or an image-only predictor can already solve, and a causal-faithfulness loss that penalises the model whenever the probability of the gold token fails to drop under image masking. We pair the recipe with IMG_correct, a refinement of the standard image-masking gap that excludes a known failure mode in which a consistency-loss-trained model flips its top prediction under masking without losing support for the right answer. On public-data OpenI chest radiographs, the method drives IMG_correct from 2.38 ± 0.26 to 74.96 ± 0.83 pp at matched token-level accuracy across three seeds. A cross-counterfactual evaluation reports honestly that the gain is largely specific to the trained counterfactual: cropping and centre occlusion collapse it. A retroactive IMG_correct audit of 18 prior verifier cells finds induced-inversion contributions uniformly small, so the methodological refinement verifies prior numbers rather than overturning them.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1. Introduction

Multimodal medical AI systems produce text that disregards the visual input: Restrepo et al. (2025) document this across six open-source chest-X-ray vision-language models with Selective Modality Shifting. The phenomenon mirrors the partial-input shortcuts of NLI (Gururangan et al., 2018; Poliak et al., 2018) and the multimodal-reward-model shortcuts of Li et al. (2025a). Two gaps motivate this work. First, existing mitigations are either measurement-only (the diagnostic literature) or frozen-predictor wrappers (conformal selection and post-hoc filtering; Gui et al., 2024; Li et al., 2025b; Mohri & Hashimoto, 2024); none modifies what the underlying generator has learned. Second, the only training-time mitigation at scale — the Shortcut-Failure Coefficient (SFC) of Li et al. (2025a) — operates only on the text-only side. We close both gaps.

Contributions. (C1) Dual-adversarial causal-faithfulness training: a LoRA fine-tune of MedGemma-4B-IT (Sellersgren et al., 2025; Hu et al., 2022) that combines a symmetric text + image adversarial filter (the image-only branch is the incremental axis relative to Li et al., 2025a) with a token-wise correctness-gain margin loss on gold tokens. (C2) IMG_correct: a methodological refinement of the image-masking gap (IMG) that excludes the induced inversion failure mode in which a consistency-loss-trained model flips its top prediction under masking without losing support for the right answer. We apply IMG_correct retroactively to prior verifier checkpoints and find induced-inversion contributions are uniformly small (Section 4). We additionally run a cross-counterfactual evaluation that honestly reports the gain is largely specific to the trained counterfactual.

2. Methods

We fine-tune MedGemma-4B-IT (Sellersgren et al., 2025) with LoRA (Hu et al., 2022) rank 16 on q/k/v/o projections (11.9 M trainable parameters; full hyperpa-

rameters in the supplementary appendix). The training objective is

$$\mathcal{L}(x) = w(x) \cdot \mathcal{L}_{\text{SFT}}(x) + \lambda_{\text{faith}} \cdot \mathcal{L}_{\text{faith}}(x), \quad (1)$$

with $\lambda_{\text{faith}} = 0.3$.

Dual adversarial filter. The per-row weight is $w(x) = \min(w_{\text{text}}(x), w_{\text{image}}(x))$, combining two shortcut detectors trained independently: a RoBERTa-large text-only cross-encoder (reusing prior verifier infrastructure [citation withheld for double-blind review]) and a BiomedCLIP (Zhang et al., 2024) image-only MLP head. Rows on which either branch’s gold-class probability exceeds 0.7 are downweighted to $w_{\text{min}} = 0.2$. On the 20,560-row training pool, the image-only branch catches 6.5% of rows the text-only branch misses — the incremental axis relative to Li et al. (2025a) (decomposition in the supplementary appendix).

Causal-faithfulness loss. The naive consistency loss $\text{ReLU}(m - \text{KL}(p_{\text{full}} \| p_{\text{alt}}))$ admits a degenerate optimum: the model can satisfy the margin by being arbitrarily wrong under masking. We replace symmetric KL with a token-wise correctness-gain margin on gold tokens,

$$\mathcal{L}_{\text{faith}}(x) = \frac{1}{|T|} \sum_{t \in T} \text{ReLU}(m - (\log p_{\text{full}}(y_t) - \log p_{\text{alt}}(y_t))), \quad (2)$$

which couples the divergence requirement to gold-token correctness on the full input ($m = 0.3$ nats). Image masking uses the per-channel mean of the batch rather than a literal zero tensor: post-normalisation zeros are out-of-distribution adversarial input, whereas the per-channel mean is the model’s neutral grey image (supp. appendix).

IMG_correct. The standard image-masking gap counts every arg max flip under masking as image-grounded behaviour. A consistency-loss-trained model can drive image-zeroed accuracy to zero by always flipping its arg max even when its gold-class probability barely drops. IMG_correct counts only examples where (a) the full input is correct, (b) the masked input is incorrect, and (c) the gold-class probability drops by at least margin τ . We report induced inversion as the per-example contribution caught by IMG but excluded by IMG_correct (formal derivation in the supplementary appendix).

3. Experimental Setup

Data. Training and evaluation use public-data sources only: OpenI (Demner-Fushman et al., 2016)

Table 1. Headline IMG_correct on the OpenI test split (token-level, $n = 33,173$). Seed-mean $\pm$ standard deviation over three seeds. Induced-inversion (Inv) contributions are essentially zero on all faith-loss configurations.

Config	IMG_corr	IMG	Inv	Acc (%)
sft_only	2.38 ± 0.26	0.36	0.40	74.31
sft_dual	2.10 ± 0.24	0.89	0.58	75.41
sft_faith	74.11 ± 0.26	74.04	0.01	75.61
sft_full	74.96 ± 0.83	74.96	0.01	75.48
sft_full_sfc	75.42 ± 0.28	75.43	0.01	75.44

(10,839 rows after text-target filtering; OpenI test split with 1,587 examples for in-distribution eval) and ChestX-Det10 silver targets (Liu et al., 2020) for cross-site eval.

Configurations. Five training configurations $\times$ three seeds: sft_only (SFT baseline), sft_dual (filter only), sft_faith (faith loss only), sft_full (both; headline configuration), and sft_full_sfc (replaces the discrete dual filter with continuous SFC weights from Li et al., 2025a).

Metrics. IMG_correct at $\tau = 0.1$; standard IMG; induced inversion; token-level accuracy on gold target tokens.

4. Results

Headline (Table 1). The pre-registered ≥ 5 pp improvement is exceeded by an order of magnitude on three seeds $\times$ three faith-loss configurations. The full method sft_full lifts IMG_correct by +72.58 pp at +1.17 pp accuracy over the SFT-only baseline. The dual-filter-only configuration is statistically indistinguishable from baseline; the faith loss is the load-bearing component (sft_faith alone gives +71.73 pp). Induced inversion is essentially zero across all faith-loss configurations.

Cross-counterfactual eval (honesty check). We evaluate sft_full (seed 42) under three counterfactuals the faith loss did not train on. Under noise replacement IMG_correct stays at 62.20 pp. Under crop (9% true-image patch retained) it collapses to 4.81 pp. Under centre occlusion (25% of the image area zeroed) it collapses to 1.11 pp. The trained-mean reference is 74.00 pp on this seed. The gap-to-baseline on crop is $4.81 - 2.80 = 2.01$ pp — an order of magnitude smaller than the headline gap on the trained counterfactual (cf. the supplementary appendix). The model tightly fits the trained per-channel-mean distribution but does not transfer to localised counterfactuals.

Cross-site transfer. On 1,122 ChestX-Det10 silver-target rows, `sft_full` attains `IMG_correct` = 54.62 pp at 54.65% accuracy; the matched SFT-only baseline attains 5.12 pp. The cross-site gap of 49.5 pp is $\approx 68\%$ of the in-distribution gap, consistent with partial transfer (supp. appendix).

`IMG_correct` retroactive audit. We apply `IMG_correct` to 18 previously reported verifier configurations (nine v5/v6 headline cells + nine v5 ablations). Induced-inversion contributions are uniformly ≤ 1.31 pp absolute ($\leq 13.2\%$ of the corresponding IMG). The pre-registered $\geq 30\%$ hypothesis is refuted on every cell measured. The `abl_no_consist` cell — a v5 variant with the consistency loss removed — collapses `IMG_correct` to 3.60 pp while also reducing induced inversion, the opposite of what the pre-registered concern would predict (full table in the supplementary appendix).

5. Discussion

What `IMG_correct` shows about prior systems. Prior IMG numbers on the v5/v6 verifier family were largely measuring real image-grounded behaviour. The `IMG_correct` contribution is therefore (a) a verification that increases confidence in prior IMG claims under a strict per-example correctness criterion, and (b) a default-to-report-alongside-IMG diagnostic that detects the failure mode if less calibrated training recipes were to introduce it.

Cross-counterfactual partial generalisation. The +72.58 pp headline measures responsiveness to the trained counterfactual, with strong transfer to similar untrained globally-uniform counterfactuals (noise) and weak transfer to localised counterfactuals (crop, occlude). The honest read: training-time counterfactual interventions can tightly fit the in-distribution counterfactual; a more diverse training mix is the natural follow-up.

Public-data only. Reproducibility in medical imaging is limited by access barriers to MIMIC-CXR-credentialed data (Johnson et al., 2019). Our public-data commitment (OpenI training + ChestX-Det10 silver cross-site) means any researcher with comparable cloud compute can re-run the pipeline. This costs scale (MIMIC-CXR has $\sim 50\times$ more rows), but the methodology generalises.

6. Limitations and Conclusion

Limitations. (L1) The faith loss and `IMG_correct` measure overlapping per-token quantities; the cross-counterfactual eval quantifies this caveat empirically. (L2) OpenI-only training; cross-site only on silver targets. (L3) Filter calibration was trained for our prior verifier task and reused; the conceptually-aligned per-task filter is an approximation. (L4) Single backbone; the retroactive audit is on our own verifier checkpoints only.

We presented dual-adversarial causal-faithfulness training and the `IMG_correct` methodological refinement. Pre-registered empirical claims, tested transparently: (i) ≥ 5 pp `IMG_correct` improvement: hit by an order of magnitude. (ii) $\geq 30\%$ of v5.3’s IMG as induced inversion: refuted (1.1%). (iii) the dual filter dominates the faith loss: inverted (faith loss is load-bearing). `IMG_correct` lands independently of the empirical results and is a reusable audit tool for any IMG-style metric. Counterfactual evidence-sensitivity testing should precede aggregate-accuracy reporting for any multimodal medical generator.

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